

NCIE-014 Skeletal v0.1 → Full v1.0

Full-Expansion Change Register

Database & Storage Engineering Specification — IN DEVELOPMENT / FOR HUMAN REVIEW

Source-Confirmed Title Verification (§0.1)

Unlike the NCIE-013 v1.0 edition (which required a pre-approval surgical correction after its original build falsely stated no Master Controlled Register was supplied), the approved NCIE-014 Skeletal v0.1 already carries a source-confirmed title ('NCIE-014 — Database & Storage Engineering Specification') and source-confirmed suite purpose in its own Document Governance block. This was checked against the NCIE Documentation Suite Master Controlled Register v2.0, Section 3, NCIE-014 entry, and found to match exactly (same title wording, same Primary Purpose framing). No title-reconciliation discrepancy exists and no correction was required for this edition.

Maps each of the 34 approved NCIE-014 Skeletal v0.1 chapters to its expanded v1.0 location, states the authority basis for the expansion, and records the effect on that chapter's Human Review Focus item. No chapter was added, removed, merged, split or renumbered; every approved chapter purpose, Salient Things to Note, principal component, upstream dependency and Human Review focus is substantively represented below, including an explicit Salient Things to Note subsection in all 34 chapters (directive item 75 hard gate).

Ch.	Approved Skeletal Component (v0.1)	Expanded Location (v1.0)	Authority Basis	Human Review Effect	Status
1	Mandate, authority chain, persistence boundary; HR focus: acceptance authority.	§1.0-1.3, Table 1 (boundary), Table 2 (Physical-Store Allocation Matrix, item 55), Table 3 (RACI), HR14-1-1.	Production-Ready register; NCIE-003/004/006/009/010/011.	HR14-1-1 confirms NCIE-014 acceptance/sign-off authority, Blocking.	EXPANDED
2	Persistence engineering doctrine; HR focus: principles/trade-offs.	§2.1-2.3, Table 4 (RACI), HR14-2-1.	NCIE-002/003/009/010.	HR14-2-1 classified Blocking.	EXPANDED
3	Canonical-to-physical mapping governance; HR focus: naming/mapping conventions.	§3.1-3.3, Table 5 Canonical-to-Physical Mapping Matrix (item 56), Table 6 (RACI), HR14-3-1.	NCIE-003 canonical dictionary/ERDs.	HR14-3-1 classified Blocking.	EXPANDED
4	Relational/transactional persistence design; HR focus: relational technology/transactions.	§4.1-4.3, Table 7 (RACI), HR14-4-1.	NCIE-003/004.	HR14-4-1 classified Blocking.	EXPANDED
5	Analytical/columnar persistence design; HR focus: analytical technology/workloads.	§5.1-5.2, Table 8 (RACI), HR14-5-1.	NCIE-002/003/013.	HR14-5-1 classified Blocking.	EXPANDED
6	Object/blob storage design; HR focus: object-store technology/sensitive classes.	§6.1-6.2, Table 9 (RACI), HR14-6-1.	NCIE-003/009/010.	HR14-6-1 classified Blocking.	EXPANDED
7	Search index engineering; HR focus: search technology/freshness.	§7.1-7.2, Table 10 (RACI), HR14-7-1.	NCIE-006/009/010.	HR14-7-1 classified Blocking.	EXPANDED
8	Vector index/embedding persistence; HR focus: vector technology/lifecycle/sensitive-data policy.	§8.1-8.2, Table 11 (RACI), HR14-8-1.	NCIE-006/009/010.	HR14-8-1 classified Blocking.	EXPANDED
9	Graph persistence & relationship projections; HR focus: graph technology/projection ownership.	§9.1-9.2, Table 12 (RACI), HR14-9-1.	NCIE-003/010.	HR14-9-1 classified ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED, Non-Blocking.	EXPANDED
10	Time-series & metrics persistence; HR focus: time-series technology/retention/rollups.	§10.1-10.2, Table 13 (RACI), HR14-10-1.	NCIE-003/004/013.	HR14-10-1 classified Blocking.	EXPANDED
11	Agent/Model/Tool/VPF registry persistence; HR focus: physical store choices/history.	§11.1-11.2, Table 14 (RACI), HR14-11-1.	NCIE-003/004/007/009.	HR14-11-1 classified Blocking.	EXPANDED

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12	Memory & Context persistence; HR focus: store allocation/retention/ephemeral policy.	§12.1-12.2, Table 15 (RACI), HR14-12-1.	NCIE-006/009.	HR14-12-1 classified Blocking.	EXPANDED
13	Evidence, provenance & audit persistence; HR focus: storage classes/assurance.	§13.1-13.2, Table 16 (RACI), HR14-13-1.	NCIE-009/010.	HR14-13-1 classified Blocking.	EXPANDED
14	Security/identity/secrets/key-metadata persistence interface; HR focus: which security records reside where.	§14.1-14.2, Table 17 (RACI), HR14-14-1.	NCIE-009 Ch.3-12.	HR14-14-1 classified Blocking.	EXPANDED
15	Physical keys, constraints & referential integrity; HR focus: constraint strictness.	§15.1-15.2, Table 18 (RACI), HR14-15-1.	NCIE-003 identifiers/cardinality.	HR14-15-1 classified ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED, Non-Blocking.	EXPANDED
16	Indexing & query access paths; HR focus: critical query/workload priorities.	§16.1-16.2, Table 19 (RACI), HR14-16-1.	NCIE-001/003/013.	HR14-16-1 classified ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED, Non-Blocking.	EXPANDED
17	Partitioning, sharding & distribution; HR focus: partition/shard criteria/sovereignty.	§17.1-17.2, Table 20 (RACI), HR14-17-1.	NCIE-002/003/009.	HR14-17-1 classified Blocking.	EXPANDED
18	Replication, consistency & authoritative-read semantics; HR focus: consistency by workload.	§18.1-18.3, Table 21 Authoritative-Read Matrix (item 57), Table 22 (RACI), HR14-18-1.	NCIE-006/009/011.	HR14-18-1 classified Blocking.	EXPANDED
19	Caching & materialized views; HR focus: cacheable classes/invalidation.	§19.1-19.2, Table 23 (RACI), HR14-19-1.	NCIE-006/009/011.	HR14-19-1 classified Blocking.	EXPANDED
20	Data classification, residency & sovereignty enforcement; HR focus: approved residency rules.	§20.1-20.3, Table 24 Residency-Copy Matrix (item 58), Table 25 (RACI), HR14-20-1.	NCIE-003/008/009.	HR14-20-1 classified Blocking.	EXPANDED
21	Encryption, tokenization & protected-field storage; HR focus: protection policy by data class.	§21.1-21.2, Table 26 (RACI), HR14-21-1.	NCIE-003/009.	HR14-21-1 classified Blocking.	EXPANDED
22	Retention, legal hold, archival & disposition engineering; HR focus: periods/hold authority/disposal.	§22.1-22.3, Table 27 Retention/Disposition Matrix (item 59), Table 28 (RACI), HR14-22-1.	NCIE-006/008/010.	HR14-22-1 classified Blocking.	EXPANDED
23	Backup, restore & PITR; HR focus: RPO/RTO/backup classes.	§23.1-23.3, Table 29 Recovery/Reconciliation Matrix (item 60), Table 30 (RACI), HR14-23-1.	NCIE-004/009/010.	HR14-23-1 classified Blocking.	EXPANDED
24	Disaster recovery & cross-site persistence; HR focus: DR topology/sovereignty.	§24.1-24.2, Table 31 (RACI), HR14-24-1.	NCIE-004/009/010.	HR14-24-1 classified Blocking.	EXPANDED
25	Schema migration & database change management; HR focus: migration approval/downtime.	§25.1-25.2, Table 32 (RACI), HR14-25-1.	NCIE-003/004/011.	HR14-25-1 classified Blocking.	EXPANDED
26	Data migration, backfill & legacy reconciliation; HR focus: legacy sources/acceptance/reconciliation authority.	§26.1-26.2, Table 33 (RACI), HR14-26-1.	NCIE-003/010.	HR14-26-1 classified Blocking.	EXPANDED
27	Data lifecycle reconciliation & repair; HR focus: auto-repair boundaries/approval.	§27.1-27.2, Table 34 (RACI), HR14-27-1.	NCIE-003/010/011.	HR14-27-1 classified Blocking.	EXPANDED

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28	Database security & privileged administration; HR focus: DBA privilege model/emergency authority.	§28.1-28.2, Table 35 (RACI), HR14-28-1.	NCIE-008/009.	HR14-28-1 classified Blocking.	EXPANDED
29	Storage observability, capacity & health; HR focus: storage SLO/capacity thresholds.	§29.1-29.2, Table 36 (RACI), HR14-29-1.	NCIE-004/010/015.	HR14-29-1 classified Blocking.	EXPANDED
30	Performance engineering & query optimization; HR focus: critical workloads/performance targets.	§30.1-30.2, Table 37 (RACI), HR14-30-1.	NCIE-002/013.	HR14-30-1 classified ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED, Non-Blocking.	EXPANDED
31	Cost, tiering & storage efficiency; HR focus: cost objectives/permitted tiering.	§31.1-31.2, Table 38 (RACI), HR14-31-1.	NCIE-004/008.	HR14-31-1 classified Blocking.	EXPANDED
32	Database/storage testing & verification; HR focus: test acceptance authority/scenarios.	§32.1-32.2, Table 39 (9 test classes, directive item 52), Table 40 (RACI), HR14-32-1.	NCIE-009/010/016.	HR14-32-1 classified Blocking.	EXPANDED
33	Human Review Register & open persistence decisions; HR focus: register/decide register items.	§33.0-33.6, Table 41 (summary), Table 42 (complete register, 34 items), HR14-33-1.	All chapters above (mechanically reconciled — directive item 61); approved Skeletal v0.1 Chapter 33 Salient Things to Note.	HR14-33-1 (this chapter's own item) included in Table 42 and totals from the start — the NCIE-008 register-reconciliation defect and NCIE-009 salient-note omission were both designed around, not repeated. All 34 items consolidated.	EXPANDED
34	Traceability, acceptance & downstream handover; HR focus: skeleton approval/expansion authorization.	§34.0-34.6, Table 43 (traceability matrix, 34 rows), Table 44 (downstream handover, 3 rows), HR14-34-1.	Approved NCIE-001-013; approved NCIE-014 Skeletal v0.1.	HR14-34-1 unchanged, RESOLVED BY UPSTREAM BASELINE (skeleton approval already satisfies it).	EXPANDED

Consolidated Matrices Added (Directive Items 55-60)

Table 2 (Chapter 1) — Physical-Store Allocation Matrix, mapping canonical/logical object classes to physical persistence roles without creating semantic ownership. Table 5 (Chapter 3) — Canonical-to-Physical Mapping Matrix. Table 21 (Chapter 18) — Authoritative-Read Matrix, fixing which workload classes require authoritative current state versus permitted stale/replica/cache reads. Table 24 (Chapter 20) — Residency-Copy Matrix, covering every copy type (primary, replica, index, cache, backup, DR, archive). Table 27 (Chapter 22) — Retention/Disposition Matrix. Table 29 (Chapter 23) — Recovery/Reconciliation Matrix, covering restore, DR failover, schema migration rollback, legacy migration exceptions and repair scenarios. Every other chapter carries its own chapter-appropriate RACI/control-ownership table.

Human Review Register Reconciliation

34 Human Review items total: 32 items from Chapters 1-32 (one per chapter, mechanically extracted from each chapter's own Human Review Focus bullet text), plus HR14-33-1 (Chapter 33's own item) and HR14-34-1 (Chapter 34's own item) — both defined once and included in the register from the start, per the lessons learned from the NCIE-008 v1.0 correction (register-reconciliation omission) and the NCIE-009 v1.0 correction (Salient Things to Note omission), rather than discovered afterward. Classification breakdown: 12 Institutional Confirmation Required, 10 Security/Sovereignty Confirmation Required, 9 Engineering/Infrastructure Discovery Required, 2 Legal/Policy Confirmation Required, 1 Resolved by Upstream Baseline. 29 Blocking / 5 Non-Blocking. Every NCIE-014 item was individually checked against the existing NCIE-008–013 Human Review registers per directive item 62's upstream-deduplication rule; none was found to restate an identical institutional/security/domain question, so no cross-reference deduplication was required for this edition. No item manufactures closure of a decision the approved skeleton, the approved NCIE-008–013 layers, or the approved upstream chain does not itself settle.

Skeleton Fidelity Statement

Every approved Chapter Purpose, Salient Things to Note, Principal Component, Upstream Dependency/Authority and Human Review Focus from the approved NCIE-014 Skeletal v0.1 is substantively represented in the corresponding v1.0 chapter above, including an explicit Salient Things to Note subsection in all 34 chapters — independently re-verified against the final assembled document by a programmatic scan (directive item 75), not assumed from the build script; none is omitted and none is conceptually inverted (Chapter 34 §34.3).